

AeroGard

AI-Powered Hyperlocal Air Quality and Personalized Respiratory Health Alert System

Progress report — Phases 1 to 3 of 6 · 25 September 2026

Final Year Project + Research Paper · Study area: Bengaluru · Diseases: Asthma, COPD

Summary

AeroGard is being built in dependency order: data first, then forecasting, exposure, risk, alerts, API and finally the Android app. Phases 1–3 are complete and verified. The backend runs, **753,481 hourly air-quality observations** from **14 official CPCB/KSPCB monitoring stations** and **13,992 hours of weather** are stored and validated, and a 24-hour forecast model is trained, evaluated against observed data, versioned and served through the API. The automated suite of **232 tests** passes.

Phase	Scope	Status
1	Foundation: database, migrations, authentication, patient profile, security	Done
2	Air-quality and weather ingestion, validation, cleaning, hyperlocal matching	Done
3	24-hour air-quality forecast, personal exposure estimate	Done
4	Synthetic patient cohort, respiratory-risk model, calibration, personalisation	Next — needs approval
5	Explainability (SHAP), alert engine, remaining API	Not started
6	Flutter Android app, end-to-end testing, research freeze	Not started

Headline results

- 24-hour $PM_{2.5}$ forecast: mean absolute error **5.78 $\mu g/m^3$** on an untouched test period, **+23%** better than the persistence baseline.
- The model beats both baselines for $PM_{2.5}$, PM_{10} and NO_2 . For O_3 the simple 'same hour yesterday' baseline has lower MAE — reported as a finding.
- Five data-quality problems were found and fixed, including a CO unit error in the source data and a data-leakage bug in our own cleaning code (Section 5).

1. Phase 1 — Foundation and security

- **Stack:** FastAPI (Python), PostgreSQL 16 in Docker, SQLAlchemy 2 with Alembic migrations.
- **Layering:** routes → services → repositories → database. Routes contain no SQL or ML (PRD §5.1).
- **Authentication:** register, login, refresh, logout. Passwords hashed with Argon2. Short-lived JWT access tokens; refresh tokens rotate on every use and a replayed (stolen) refresh token revokes all of that user's sessions.
- **Authorisation:** patient-scoped endpoints check ownership; another patient's record returns 404 (same as 'not found') so IDs cannot be probed. Tested explicitly: user A cannot read user B's data.
- **Patient profile:** age, sex, disease type (asthma / COPD), severity, consent status, baseline information. Clinical fields stay empty until provided — never defaulted.
- **Audit log** for sensitive actions (login, consent change, profile update) — records field names and counts, never health or location values.
- **Error contract:** every error is `{code, message, request_id}`; no stack traces reach the client. Structured JSON logs with a request ID per call.

2. Phase 2 — Environmental data pipeline

2.1 Sources

Source	Used for	Status
OpenAQ v3 (CPCB/KSPCB reference monitors)	Ground-truth hourly air quality, history + recent	Active — 14 stations with data (20 station records)
Open-Meteo ERA5 reanalysis	Historical hourly weather	Active
Open-Meteo forecast	Weather for live forecasts	Active
data.gov.in (CPCB live feed)	Optional real-time fallback	Built, disabled: portal returned HTTP 502; units unverified
Low-cost sensors (e.g. AirGradient)	—	Excluded from ground truth by design

2.2 Data loaded (Bengaluru)

Pollutant	Stations	Hourly rows	% flagged	Median $\mu\text{g}/\text{m}^3$	Period
SO ₂	14	133,264	13.1%	19.0	Feb 2025 – Sep 2026
PM ₁₀	14	131,429	11.7%	68.7	Feb 2025 – Sep 2026
NO ₂	14	130,878	12.0%	32.9	Feb 2025 – Sep 2026
CO	14	129,348	15.0%	600.0	Feb 2025 – Sep 2026
O ₃	12	114,773	12.5%	23.7	Feb 2025 – Sep 2026
PM _{2.5}	12	113,789	12.3%	25.3	Feb 2025 – Sep 2026

Quality flags: valid 657,187; low coverage (<75% of readings) 95,379; provider-flagged 914; implausible 1. Flagged records are kept and labelled, not deleted.

2.3 Processing rules

- **Validation:** records with missing/naive timestamps, unknown pollutants, invalid coordinates, missing, non-finite or negative values, or unknown units are rejected and the reason counted.
- **Units:** everything stored in $\mu\text{g}/\text{m}^3$; gases in ppb converted at 25 °C / 1 atm ($\mu\text{g}/\text{m}^3 = \text{ppb} \times \text{molecular weight} / 24.45$). Unknown conversions are rejected, never guessed.
- **Timestamps:** UTC, stamped at the end of the averaging hour, so a value is fully known at its timestamp.
- **Deduplication:** one record per (source, station, pollutant, hour); re-loading is idempotent.

- **Cleaning:** hourly grid without invented values; model inputs fill gaps of ≤ 3 h using past data only; outliers flagged with a trailing robust z-score.
- **Hyperlocal matching:** haversine distance to stations within 10 km, stale (>3 h) or future readings dropped, inverse-distance weighting over the 4 nearest; an explicit 'unavailable' with a reason when nothing usable is nearby.

3. Phase 3 — 24-hour forecast and exposure

3.1 Forecast method

- **Features (35):** current value, lags (1, 6, 12, 24, 48 h), rolling mean/max/std, other pollutants, weather, calendar (IST), station location. Every feature uses only data at or before the prediction time — enforced by automated leakage tests.
- **Models:** persistence and seasonal-naive baselines; LightGBM with one model per horizon (1–24 h).
- **Protocol:** chronological split, never shuffled, 24 h embargo at each boundary. Train Feb 2025–Jan 2026, validation Feb–Apr 2026 (early stopping only), test May–Sep 2026 (used once). All models scored on the same forecast–observation pairs, against observed values only.
- **Artifacts:** each model saved with a metadata file (versions, periods, metrics, SHA-256). The API refuses a model whose checksum or feature list does not match.

3.2 Results on the test period (valid hours, error in $\mu\text{g}/\text{m}^3$)

Pollutant	Model	MAE	RMSE	R ²	MAE 1h	MAE 24h	Skill
PM_{2.5}	LightGBM (24 direct models)	5.78	15.67	0.31	3.18	6.39	+23%
PM _{2.5}	Seasonal naive (same hour yesterday)	6.92	22.96	-0.49	7.24	7.02	+8%
PM _{2.5}	Persistence (no change)	7.51	22.82	-0.47	3.32	7.02	+0%
PM₁₀	LightGBM (24 direct models)	14.01	26.12	0.56	6.68	15.15	+26%
PM ₁₀	Seasonal naive (same hour yesterday)	15.61	34.34	0.24	15.72	15.66	+18%
PM ₁₀	Persistence (no change)	19.03	38.80	0.03	7.32	15.66	+0%
NO₂	LightGBM (24 direct models)	5.74	37.69	0.24	3.76	6.08	+28%
NO ₂	Seasonal naive (same hour yesterday)	6.83	45.79	-0.13	6.89	6.95	+14%
NO ₂	Persistence (no change)	7.93	45.39	-0.11	2.36	6.95	+0%
O₃	Seasonal naive (same hour yesterday)	4.83	11.32	0.52	4.90	4.85	+28%
O ₃	LightGBM (24 direct models)	5.02	10.51	0.59	2.79	5.29	+25%
O ₃	Persistence (no change)	6.68	14.53	0.21	3.04	4.85	+0%

Bold = lowest MAE for that pollutant. Skill = improvement in MAE over persistence.

Findings

- LightGBM clearly improves on both baselines for PM_{2.5}, PM₁₀ and NO₂, with the largest gains 6–24 hours ahead.
- **O₃:** 'same hour yesterday' has the lowest MAE; LightGBM has lower RMSE and higher R². Ozone's strong daily sunlight cycle makes the naive baseline hard to beat.
- **NO₂ at 1 hour ahead:** persistence beats LightGBM.
- **PM_{2.5}:** RMSE much larger than MAE and modest R² — the model improves typical hours but does not anticipate sudden spikes.
- The conclusion holds when flagged hours are included in scoring (secondary evaluation).

3.3 Personal exposure estimate

Exposure = air quality at the user's location × a factor for where the user is. Factors are literature-based assumptions (to be tested by sensitivity analysis), not measurements:

Setting	PM _{2.5}	PM ₁₀	NO ₂	O ₃	CO
Outdoors / unknown	1.0	1.0	1.0	1.0	1.0
Indoors at home	0.8	0.6	0.6	0.3	0.9
Indoors elsewhere	0.6	0.5	0.5	0.2	0.8
In traffic	1.3	1.3	1.6	0.8	1.8

- Hours without a location fix in the previous 2 hours are reported as missing, never guessed.
- 6 / 24 / 72-hour averages, maxima and cumulative exposure, each with a coverage percentage.
- Location and exposure require the patient's consent.

4. API available now

Endpoint	Purpose	Access
GET /health	Service and database status	Public
POST /auth/register · login · refresh · logout	Accounts and sessions	Public / signed in
GET, PATCH /patients/me	Own patient profile and consent	Signed in
GET /air/current	Location-matched air quality (optional past time)	Signed in
GET /air/forecast	24 hourly forecast values for a location	Signed in
GET /air/stations	Reference monitoring stations	Signed in
POST /location	Upload location fixes	Signed in + consent
GET /exposure/current · history	Estimated personal exposure	Signed in + consent

5. Problems found and fixed

#	Problem	Fix
1	CO on the CPCB feed labelled 'ppb' but actually mg/m ³ — values 1,000× too small. Confirmed by comparing with the older feed (1,260 vs 0.49).	Documented source correction; 129,348 stored rows converted.
2	NO ₂ and SO ₂ 'ppb' labels may also be wrong (possible 1.9× / 2.6× overstatement); evidence ambiguous.	Marked unverified; excluded from risk thresholds until checked against CPCB's own values.
3	Isolated PM _{2.5} spikes of 600–900 µg/m ³ in the monsoon, mostly low-coverage — likely sensor glitches.	Model trained and primarily scored on fully valid hours; results also reported on all hours.
4	Gap-filling by interpolation used the value after the gap — a data-leakage bug in our own code.	Model inputs now use past-only filling; tests prove future changes cannot alter inputs.
5	OpenAQ timed out (HTTP 408) on long queries; two sensors missed.	Requests split into 90-day chunks, 408 retried; both stations reloaded.

6. Testing

232 automated tests run against a dedicated test database built through the real migrations. They cover authentication and token abuse, patient isolation, the error format, source adapters (valid, empty, malformed, timeout, rate limit, partial failure), validation and unit conversion, deduplication and idempotent reloads, spatial matching edge cases, cleaning, leakage guards, forecast horizons and artifact checks, exposure, consent and the API end to end. No test calls a live external service.

7. Limitations to state in the paper

- Personal exposure is a model-based estimate; no personal sensor is used.
- All Bengaluru stations have a data gap from about Oct 2022 to Feb 2025; modelling uses Feb 2025 onward (~19 months).
- The test period covers the monsoon season only; winter performance is measured on validation only.
- OpenAQ lags real time by about 3 days, so the live 'current' view is usually reported as stale; demos use a past time.
- Weather for training is ERA5 reanalysis; live forecasts use numerical weather forecasts instead.
- Respiratory outcome labels in Phase 4 will be synthetic and must be reported as such — not clinical validation.

8. Next: Phase 4 — decisions needed

Phase 4 builds the synthetic patient cohort and the respiratory-risk model. Two proposed defaults need approval:

Decision	Proposed default
Event to predict	Exacerbation in the next 24 hours (symptom flare needing extra rescue-inhaler use), generated synthetically from exposure, severity and personal sensitivity using published effect sizes; clearly labelled synthetic.
Risk levels	Low < 15% · Moderate 15–35% · High 35–60% · Severe ≥ 60% (stored as versioned configuration, changeable without retraining).

Before Phase 6: install Flutter SDK and Android Studio; an Android phone is recommended for Health Connect and GPS testing.